

E-MOSFET AS A DIGITAL SWITCH

E-MOSFET APPLICATION

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TOPIC OUTLINE

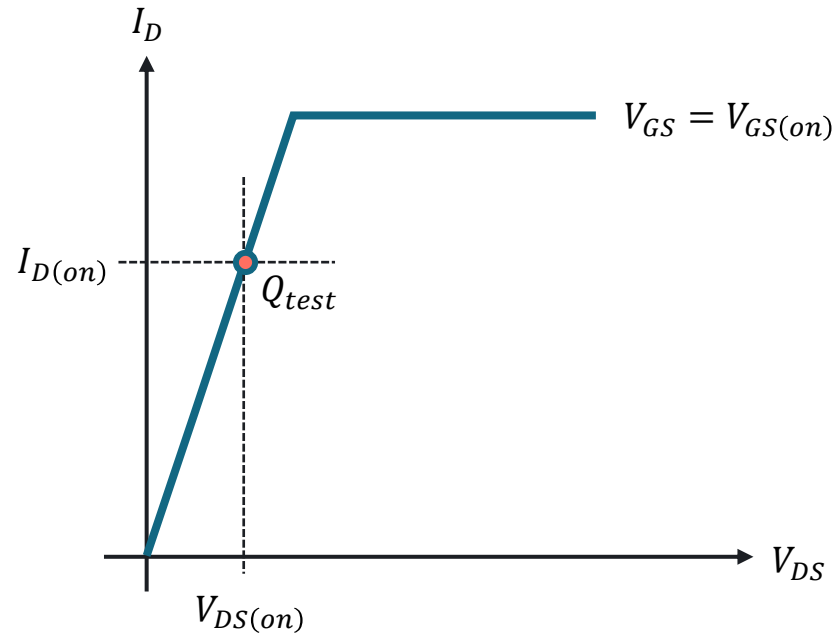
The Ohmic Region

Active-load Switching



OHMIC REGION

Measuring $R_{DS(on)}$



Drain-Source ON Resistance

$$R_{DS(on)} = \frac{V_{DS(on)}}{I_{D(on)}}$$

For instance, at the test point, a VN2406L has $V_{DS(on)} = 1V$ and $I_{D(on)} = 100mA$.

$$R_{DS(on)} = \frac{1V}{100\text{ mA}}$$

$$R_{DS(on)} = 10\Omega$$

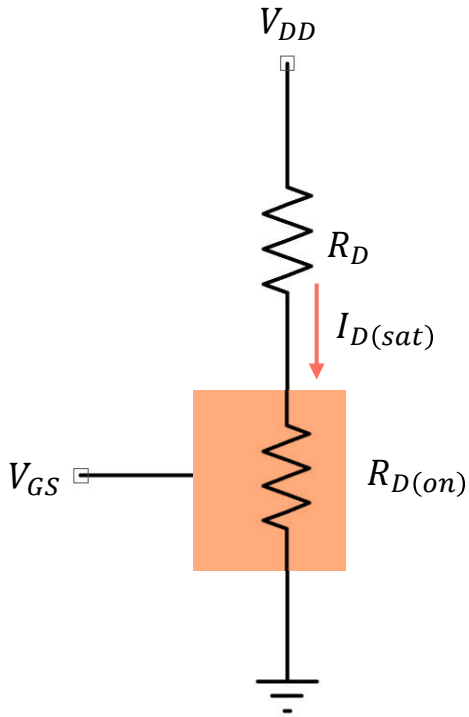


SMALL-SIGNAL E-MOSFETs

Device	$V_{GS(th)}$	$V_{GS(on)}$	$I_{D(on)}$	$R_{DS(on)}$	$I_{D(max)}$	$P_{D(max)}$
VN2406L	1.5V	2.5V	100 mA	10 Ω	200mA	350mW
BS107	1.75V	2.6V	20 mA	28 Ω	250mA	350mW
2N7000	2V	4.5V	75 mA	6 Ω	200mA	350mW
VN10LM	2.5V	5V	200 mA	7.5 Ω	300mA	1W
MPF930	2.5V	10V	1 A	0.9 Ω	2A	1W
IRFD120	3V	10V	600 mA	0.3 Ω	1.3A	1W

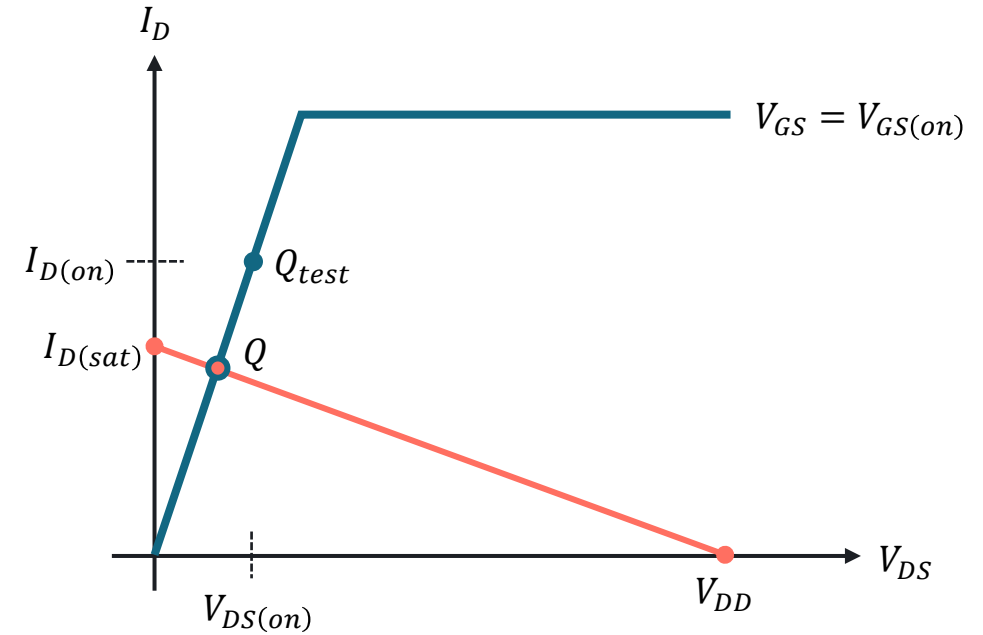


BIASING IN OHMIC REGION



Drain saturation current

$$i_{D(sat)} = \frac{V_{DD}}{R_D}$$



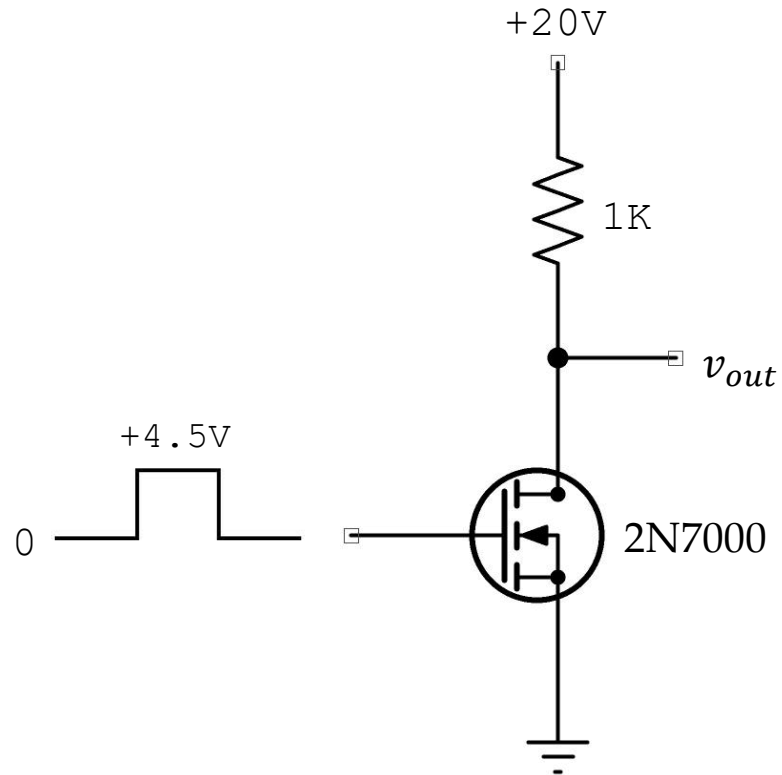
$I_{D(sat)}$ less than $I_{D(on)}$ with $V_{GS} = V_{GS(on)}$ ensures saturation.



EXERCISE

What is the output voltage in the given circuit?

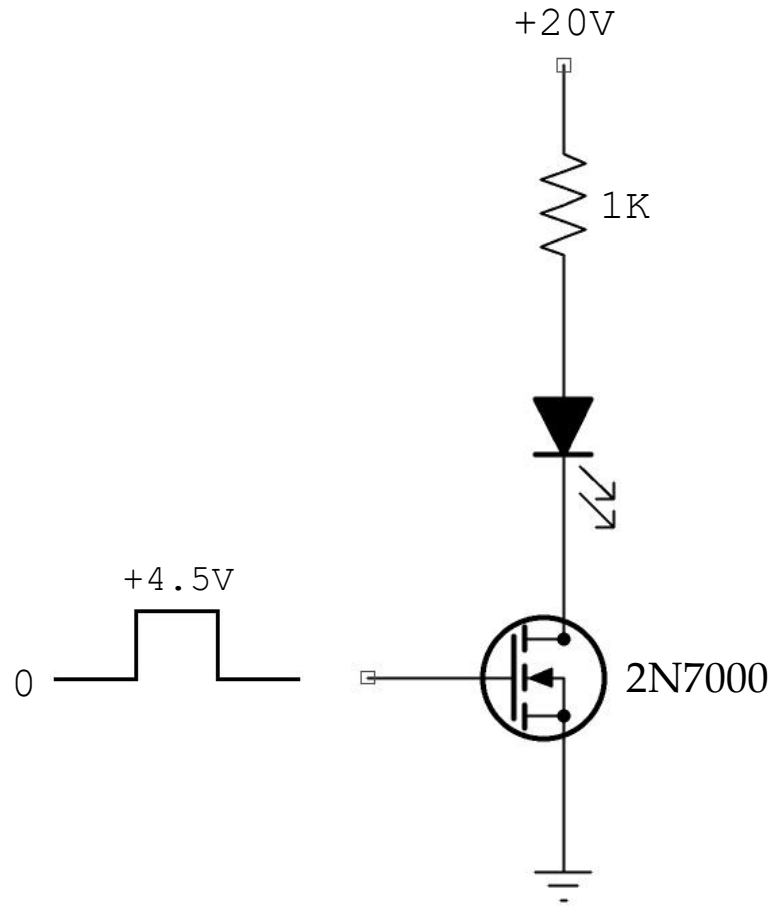
Solution



EXERCISE

Calculate the LED current in the given circuit.

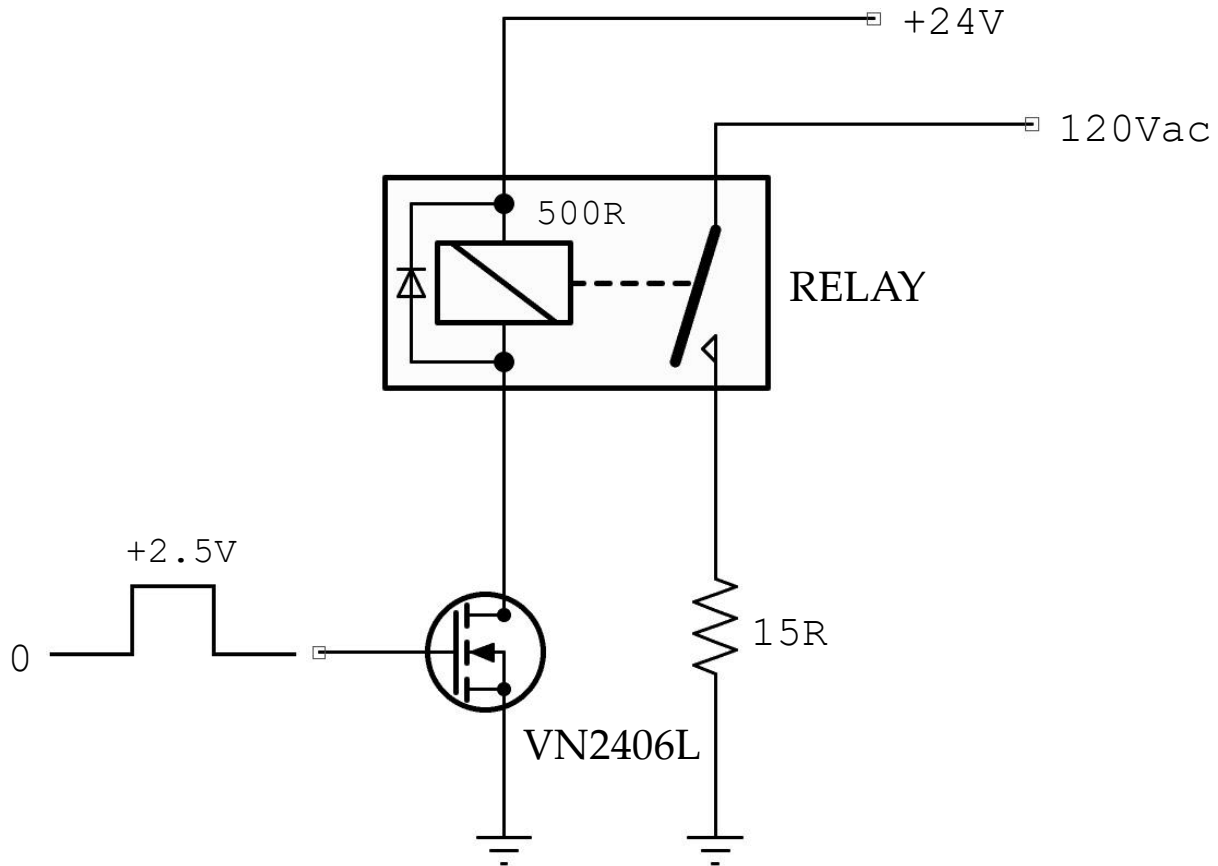
Solution



EXERCISE

What does the given circuit do if a coil current of 30mA or more closes the relay contact?

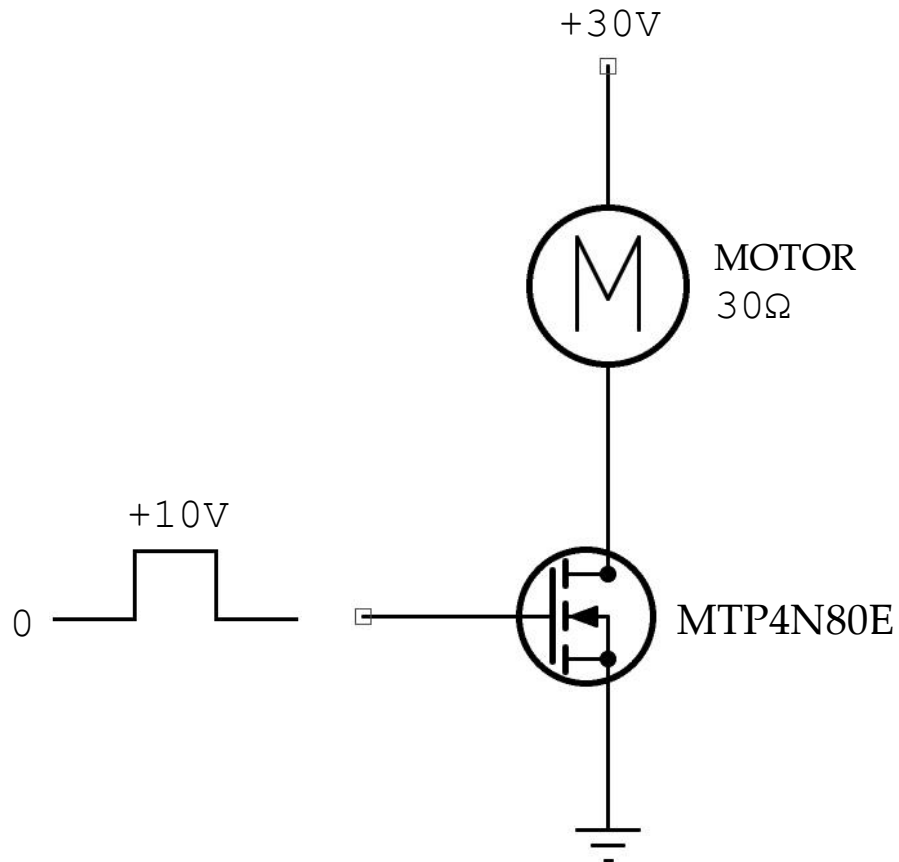
Solution



EXERCISE

What is the current through the motor winding? For an MTP4N80E: $V_{GS(on)} = 10\text{ V}$, $I_{D(on)} = 2\text{ A}$, and $R_{DS(on)} = 1.95\Omega$.

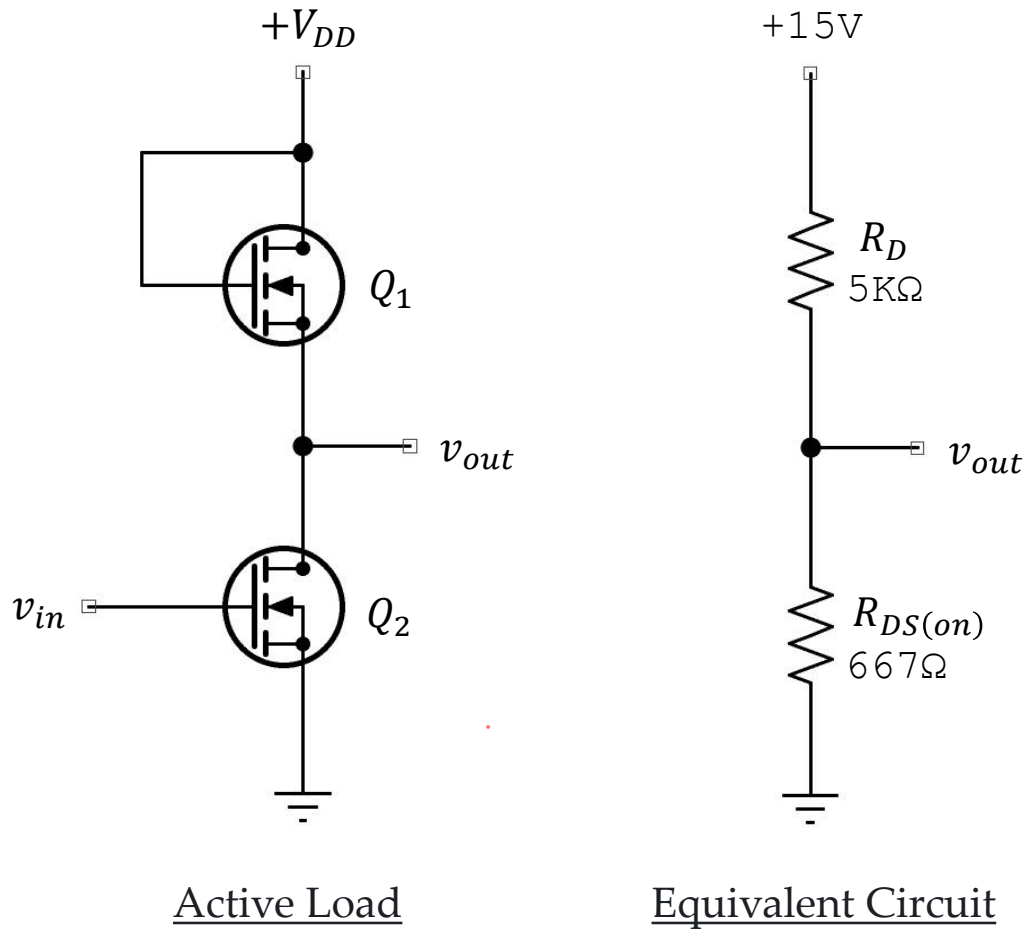
Solution



ACTIVE-LOAD SWITCHING

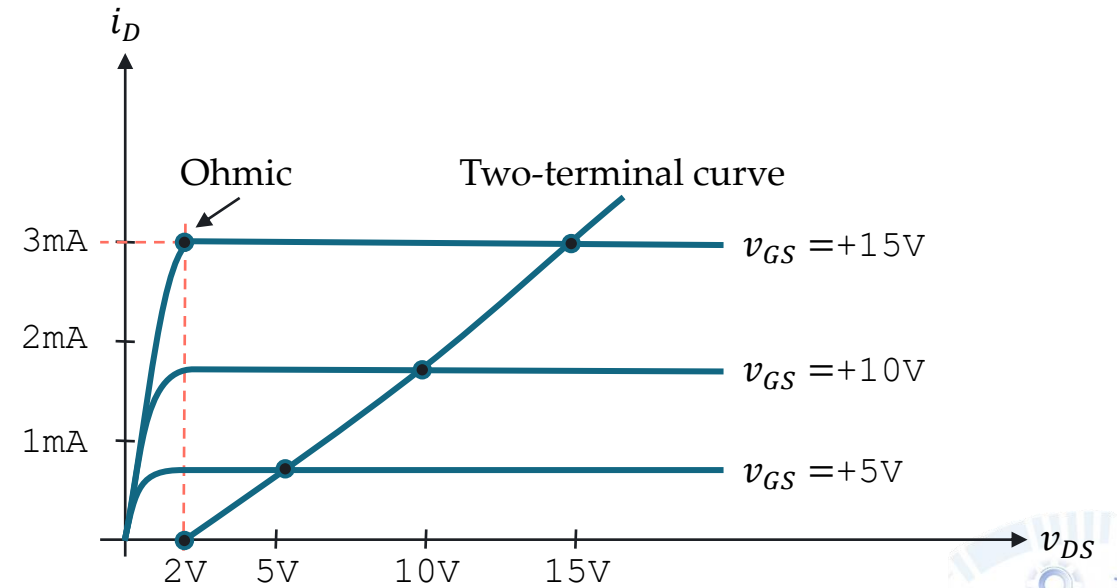


ACTIVE-LOAD SWITCHING



$V_{GS} = V_{DS}$ produces a two-terminal curve with an active resistance of:

$$R_D = \frac{V_{DS(active)}}{I_{D(active)}}$$

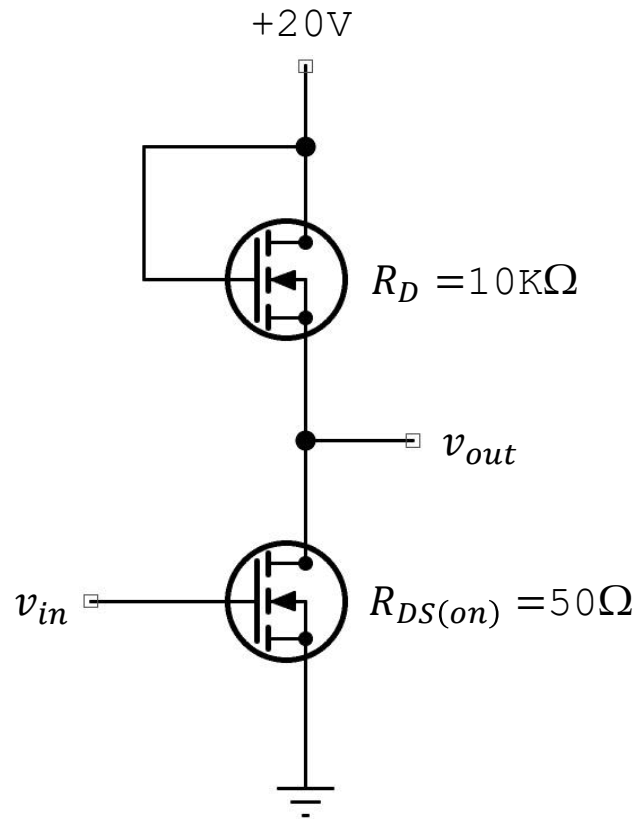


EXERCISE

What is the output voltage when the input is low?

Solution

When the input is high?

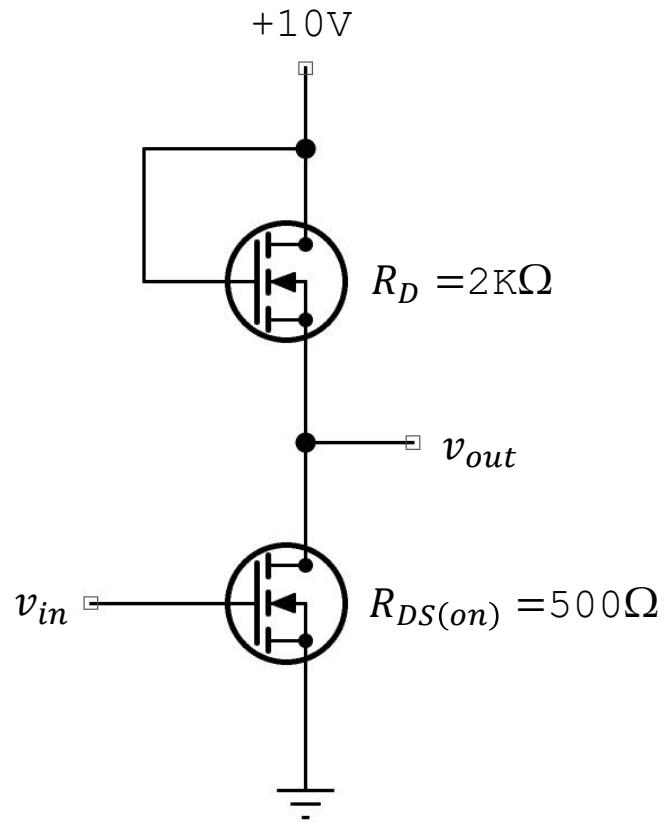


EXERCISE

What is the output voltage when the input is low?

Solution

When the input is high?



LABORATORY

